

TradePanel

Agent Handover — Live Readiness Sprint

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This document is the canonical handover for any agent or session continuing work on TradePanel. Read this before touching any code. Everything in here was verified against the live codebase on 2026-05-06.

1 System Overview

TradePanel is a full-stack algorithmic trading system. It connects MetaTrader 5 to a Python paper-trading engine, stores everything in PostgreSQL, and exposes a React dashboard + Telegram bot for monitoring and control.

File	Purpose
Root	F:\REPOS\leo123xxx\TradePanel
App folder	F:\REPOS\leo123xxx\TradePanel
Web app (FastAPI + React)	webapp/
Paper trading engine	forward_test/paper_engine.py
Telegram bot	notifications/telegram_bot.py
Command router	notifications/router.py
Notification templates	notifications/templates.py
DB schema	db/init/01_schema.sql
Scheduler	scheduler/docker_jobs.py
Strategies config	config/strategies.yaml
System config	config/config.yaml
Skills reference	TRADEPANEL_SKILLS.md (read at session start)

Stack

- Python 3.12 + FastAPI + PostgreSQL 16 + React/Vite
- MT5 bridge via MetaTrader5 Python library (Docker mock available)
- APScheduler for overnight jobs, Telegram python-telegram-bot v20
- Docker: 7-service compose stack (db, backend, frontend, telegram-bot, scheduler, waha, adminer)
- Account currency: ZAR (South African Rand) on Exness MT5 DEMO, R50 000 balance

2 Changes Made This Session (2026-05-06)

All changes are committed to the working directory. No Docker rebuild required for backend changes — `uvicorn auto-reloads`. Frontend requires `npm run build` after `App.jsx` edits.

Status	Item	Detail / Owner
DONE	New <code>/api/health</code> endpoint (<code>router_health.py</code>)	Real PostgreSQL + MT5 + Event Bus checks. Polls <code>bot_health</code> table for MT5 heartbeat.
DONE	Register health router in <code>webapp/main.py</code>	Added <code>health_router</code> import and <code>include_router</code> call.
DONE	Fix profit → <code>net_pnl</code> in <code>router_papertrades.py</code>	<code>profit</code> column does not exist in schema. Fixed summary + trades queries. Both now use <code>net_pnl</code> .
DONE	Add <code>signal_taken</code> flag to <code>/papertrades/signals</code> endpoint	<code>triggered_trade_id</code> IS NOT NULL returned as <code>signal_taken</code> boolean on every signal row.
DONE	Fix P&L in <code>router_accounts.py</code> (all 4 endpoints)	<code>COALESCE(net_pnl, exit_price - entry_price)</code> replaces raw price diff. <code>direction</code> + <code>status</code> added to trade list.
DONE	Dashboard connectivity card — remove hardcodes	<code>useConnectivity</code> hook polls <code>/api/health</code> every 15s. <code>ConnStatus</code> component colours by real value.
DONE	Signals table — add Taken column	<code>App.jsx</code> signals table: 8th column shows ☒ when <code>signal_taken=true</code> , dash otherwise.
DONE	Telegram <code>/signals</code> — show taken status per signal	<code>router.py get_signals()</code> : fetches <code>triggered_trade_id</code> , appends ☒ TAKEN / ⌚ Not taken line per signal.

3 Immediate Bugs to Fix Next Session

These were uncovered during today's work and must be fixed before the dashboard shows meaningful numbers.

3.1 `net_pnl` Never Written on Trade Close

CRITICAL All trade P&L on the dashboard is still raw pip diff

File: `forward_test/paper_engine.py`, function `_log_trade_close()` (approx line 580)

Problem: The UPDATE on trade close only sets `exit_price`, `close_time`, `status`, `close_reason`. `net_pnl` is never written, so the COALESCE fallback always fires. For FX/Gold, `exit_price - entry_price` is a meaningless number (raw price points, not ZAR).

Fix

- After MT5 closes the position, fetch the deal profit from `mt5.history_deals_get(position=ticket)`
- Write `net_pnl` = sum of deal profits for that ticket (already in account currency)
- Add `net_pnl` to the UPDATE statement alongside `exit_price`

Example SQL fix:

```
UPDATE trades SET exit_price=%s, close_time=%s, status=%s, close_reason=%s, net_pnl=%s WHERE mt5_ticket=%s
```

3.2 triggered_trade_id Never Populated

CRITICAL signal_taken always shows false — column is always NULL

File: forward_test/paper_engine.py, function _log_trade_open() (approx line 510)

Problem: When a trade opens from a signal, the code inserts into trades but never updates signals.triggered_trade_id. The link between signal and trade is never made, so signal_taken is always false in both the dashboard and Telegram.

Fix

- In _log_trade_open(), after the INSERT into trades, add a single UPDATE:

```
UPDATE signals SET triggered_trade_id = %s WHERE strategy_id = %s AND pair = %s AND
triggered_trade_id IS NULL ORDER BY timestamp DESC LIMIT 1
```

- Pass (trade_id, strategy_id, symbol) as params
- The LIMIT 1 ensures only the most recent un-linked signal for that strategy+pair gets claimed

3.3 Account Profiles Currency = USD (Should Be ZAR)

LOW Dashboard labels show USD, account is ZAR

The DB seed in 01_schema.sql inserts Demo/Live/Paper accounts with currency = 'USD'. Leo's Exness account is ZAR.

Fix

- Run: UPDATE account_profiles SET currency = 'ZAR' WHERE account_type IN ('DEMO', 'PAPER');
- Live account: update when actually going live to avoid confusion

4 Live Readiness Sprint — Phase Status

Handover doc from 2026-05-03 session. Phases 1–8. Demo account: R50 000 ZAR on Exness MT5 (DEMO).
Target risk: 1% = R500 max loss per trade.

Status	Item	Detail / Owner
PENDING	Phase 1 — Circuit Breaker	20% DD hard stop, magic number filter, /pause /resume verify, restart reconciliation
PENDING	Phase 2 — ATR Lot Sizing (1% risk)	risk_per_trade_pct: 10.0 → 1.0, wire live account balance into risk_manager
PENDING	Phase 3 — Breakeven + Manual Trade	Breakeven SL at 1:1 RR, /trade + /close_manual end-to-end test, news blackout verify
PENDING	Phase 4 — Enable >60% WR Strategies	Enable gold_momentum_breakout, rsi_pullback, ma_crossover, rsi_extremes_scalp in strategies.yaml
PENDING	Phase 5 — E2E Validation Script	17-test pipeline: pair coverage, lot sizing, SL/TP levels, signal-to-trade flow
PARTIAL	Phase 6 — Dashboard Live Data	Done: signals taken, connectivity, net_pnl. Remaining: live account equity, risk metrics panel
PENDING	Phase 8 — 48-Hour Demo Run	Acceptance: <12% DD, ≥60% WR, no runaway positions, morning brief firing at 08:00 SAST

Phase 1 — Circuit Breaker (Detail)

- Hard stop: if account equity drops by 20% from peak, pause all new entries and send Telegram alert
- Location to add: paper_engine.py — check at the top of each strategy scan cycle
- Magic number filter: _get_bot_positions() already filters by magic. Verify it is correct for all active strategies
- Restart reconciliation: on engine start, call positions_get() to re-adopt any open bot positions into the trades table (skeleton exists at line ~450)
- /pause and /resume: router.pause_bot() and router.resume_bot() exist but need a shared pause_flag read by paper_engine.py's main loop

Phase 2 — ATR Lot Sizing (Detail)

- risk_per_trade_pct is currently 10.0 in config — drop to 1.0 before any live-adjacent run
- risk_manager.calculate_lot_size(symbol, sl_points) uses account_balance from MT5. Verify it reads live balance, not a hardcoded figure
- Formula: $\text{lot_size} = (\text{balance} * 0.01) / (\text{sl_points} * \text{point_value})$. Verify point_value is correct for XAUUSD (100 oz per lot) vs FX (100,000 units)

Phase 4 — Strategies to Enable (Detail)

From the 2026-05-01 backtest — pairs with >60% WR confirmed:

Strategy	Pair	WR	Note
gold_momentum_breakout	XAUUSD H4	64%	72 trades — most significant
rsi_pullback	XAUUSD H4	60%	55 trades

rsi_pullback	GBPJPY H4	80%	Enable
rsi_pullback	USOIL H4	100%	Monitor only — small sample
ma_crossover	USDJPY H4	67%	18 trades
ma_crossover	GBPUSD H4	64%	11 trades
ma_crossover	EURUSD D1	80%	Enable
rsi_extremes_scalp	USOIL M15	75%	8 trades — small sample
bb_squeeze_scalp	USDJPY M15	100%	2 trades — monitor only

5 Critical Patterns & Gotchas

Read these before editing any .py file. Failure to follow these rules has caused silent corruption in past sessions.

5.1 Never Use Edit Tool on .py Files

RULE The Edit tool silently introduces UTF-16 null bytes into Python files

Python on Windows exits silently. Linux Python says "cannot contain null bytes". Either way, the file is broken with no error.

Correct pattern for all .py edits:

```
python3 << 'EOF' with open(path,'rb') as f: src = f.read().replace(b'\x00',b'').decode() src = src.replace(old, new) with open(path,'w') as f: f.write(src) EOF
```

- After every .py write, verify with Read tool or grep for the changed string
- To detect null bytes: `python3 -c "open(path,'rb').read().count(b'\x00')"`

5.2 DB Timestamps Are SAST, Not UTC

All data in PostgreSQL is stored in SAST (UTC+2) because MT5 bridge uses `datetime.fromtimestamp()` on Leo's Windows machine. Session windows in `base_strategy.py` are defined in UTC. A 2-hour mismatch kills all session-filtered signals.

- When writing any time comparison: confirm which timezone each side is in before comparing
- Affected strategies: ORB, `vwap_momentum`, `bb_squeeze_scalp`, `rsi_extremes_scalp` (all have session filters)

5.3 Backtest Engine Ignores `strategies.yaml` Params

The backtest engine instantiates strategy classes with no arguments: `strat_class()`. Only the hardcoded `__init__` defaults are used. YAML parameter values are ignored during backtests.

- To change backtest behaviour: edit the strategy `__init__` defaults directly
- YAML params are used by the live/paper engine only

5.4 `use_partial_tp=True` Requires ~80% WR

The default `use_partial_tp=True` splits the position at TP1, which requires a very high WR to be profitable at 2:1 RR. All crypto strategies must set `use_partial_tp=False`. This is why `multi_ema_crypto_scalper`, `silver_bullet_crypto`, and `power_of_3_and` were disabled.

5.5 YAML Files Have Null Bytes

The `strategies.yaml` file has null bytes on the Linux mount. Copy to `/tmp` and strip before parsing:

```
with open(path,'rb') as f: data = f.read().replace(b'\x00',b'') import yaml; cfg = yaml.safe_load(data)
```

5.6 Scheduler Canonical File

- Edit `scheduler/docker_jobs.py` ONLY. `scheduler/jobs.py` is a thin shim and will be overwritten.
- Cowork scheduled task `daily-tradepanel-automation` fires at 02:04 SAST

6 Database Schema Quick Reference

File	Purpose
<code>account_profiles</code>	<code>account_id</code> , <code>account_name</code> , <code>account_type</code> (DEMO/LIVE/PAPER), <code>currency</code> , <code>initial_balance</code>
<code>trades</code>	<code>trade_id</code> (UUID), <code>strategy_id</code> , <code>account_id</code> , <code>mode</code> , <code>pair</code> , <code>direction</code> , <code>lot_size</code> , <code>entry_price</code> , <code>exit_price</code> , <code>net_pnl</code> , <code>gross_pnl</code> ,

	sl_price, tp_price, open_time, close_time, status (OPENED/CLOSED), mt5_ticket
signals	signal_id, strategy_id, timestamp, pair, direction, timeframe, validity_window, indicator_values (JSONB), triggered_trade_id (UUID — NULL until trade opens)
strategies	strategy_id, name, category, status, parameters (JSONB), version
bot_health	event_type (HEARTBEAT/ERROR), status, message, meta_data
signal_outcomes	signal_id, outcome (TP1/TP2/TP3/SL/EXPIRED), bars_to_close, pnl_pips
backtest_runs	run_id, strategy_name, win_rate, sharpe_ratio, status (PASS/FAIL/REVIEW)
daily_summary	date, total_pnl, win_rate, trade_count

NOTE net_pnl is populated on trade close. triggered_trade_id is set when a trade opens from a signal. Both were NULL for all rows as of 2026-05-06 — see Section 3 for fixes.

7 Strategy Status Snapshot (as of 2026-05-01 Backtest)

Strategy	Best Pair	Best WR	Status
gold_momentum_breakout	XAUUSD H4	64%	PASS — Enable (Phase 4)
rsi_pullback	GBPJPY H4	80%	PASS — Enable (Phase 4)
ma_crossover	USDJPY H4	67%	PASS — Enable (Phase 4)
rsi_extremes_scalp	USOIL M15	75%	PASS — Small sample, monitor
bb_squeeze_scalp	USDJPY M15	100%	PASS — 2 trades only, monitor
rvgi_cci_confluence	Multiple	55%	REVIEW — Below 60% threshold
turtle_soup	EURUSD H4	52%	REVIEW
hikkake_trap	GBPUSD H4	50%	REVIEW
multi_ema_crypto_scalper	BTCUSD H4	44%	DISABLED — EMA late entry
silver_bullet_crypto	BTCUSD H4	42%	DISABLED — ICT fails in trend
power_of_3_amd	BTCUSD H4	40%	DISABLED — AMD fails in trend

8 Next Session — Prioritised Checklist

P0 — Fix Before Any Live-Adjacent Work

- Fix net_pnl on trade close (paper_engine.py _log_trade_close) — Section 3.1
- Fix triggered_trade_id on trade open (paper_engine.py _log_trade_open) — Section 3.2
- Fix account currency to ZAR in DB — Section 3.3

P1 — Live Readiness

- Phase 1: Circuit breaker 20% DD hard stop + pause_flag shared state
- Phase 2: Drop risk_per_trade_pct to 1.0, verify lot sizing formula for XAUUSD
- Phase 4: Enable gold_momentum_breakout + rsi_pullback GBPJPY/H4 in strategies.yaml

P2 — Complete & Validate

- Phase 3: End-to-end test /trade and /close_manual commands
- Phase 5: Write 17-test E2E validation script
- Phase 6: Dashboard live account equity panel, risk metrics card
- Phase 8: 48-hour demo run with acceptance criteria monitoring

P3 — Deferred (Not Blocking Live)

- Re-run overnight backtest after strategy v3 upgrades (all 173 combos)
- Pull stock CFD history (NVDA/AMD/MSFT/AAPL) before backtest
- Re-enable crypto strategies after adding D1 regime filter (ADX gate)

End of handover. All code changes from this session are saved in F:\REPOS\leo123xxx\TradePanel\TradePanel. Restart uvicorn (backend) and rebuild frontend (npm run build) to pick up changes.